MIL-C-17112A (OS)
1 February 1973
SUPERSEDING
MIL-C-17112 (Wep)
16 April 1962

MILITARY SPECIFICATION

COPPER-NICKEL-ZINC ALLOY (NICKEL-SILVER): CASTINGS

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers castings used for graduated sight drums on fire control instruments or castings used for other applications where the properties of the material as covered herein may prove desirable.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-781	Strapping, Flat, Steel
PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Box, Wood, Wirebound
PPP-B-591	Box, Fiberboard, Wood-Cleated
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner

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PPP-B-636	Box, Fiberboard
PPP-C-650	Crate, Wood, Open and Covered

Military

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies)
FED-STD-151	Metals; Test Methods

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch × 40 Inch Pallets, Skids, Runners, or Pallet Type Base

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM E 8	Tension Tests

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

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3. REQUIREMENTS

3.1 Material. The raw materials used shall be such as to produce castings in full conformance with the requirements of this specification.

3.2 Chemical composition. Castings shall conform to the chemical composition shown in table I. An analysis of each melt of castings shall be furnished by the contractor, showing the percentages of the elements designated.

*Table I

CHEMICAL COMPOSITION
(Copper Alloy No. 976)

Element	Percent
Copper	63.0 - 67.0
Tin	3.5 - 4.5
Lead	3.0 - 5.0
Zinc	Remainder
Iron	1.5 max.
Nickel	19.5 - 21.5
Manganese	1.0 max.

*3.3 Mechanical properties. Castings shall conform to the following requirements:

Tensile strength	30.0 ksi, min.
Yield strength (0.5% extension under load)	17.0 ksi, min.
Elongation in 2 in.	8%, min.

3.4 Cleaning. Castings shall be smooth and well cleaned by sand blasting, tumbling, chipping, or other process approved by the procuring activity, before inspection.

3.5 Repairing of defects.

3.5.1 Castings shall not be repaired, plugged, or welded without permission from the Government inspector. Such permission will be given only when the defects are small and do not adversely affect the strength,

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use, or machinability of the castings. Granting of permission by the procuring activity to repair defects shall not relieve the contractor of his responsibility for meeting all the requirements of the drawings and the specification.

3.5.2 Each repair shall be encircled with a ring of white paint prior to shipment.

3.6 Dimensions.

3.6.1 The responsiblity for furnishing castings, within the specified tolerances, as shown on the blueprints or drawings, and that will conform to such gages as may be specified in individual cases, shall rest with the contractor. Sufficient stock shall be allowed, where required, for finishing, but castings of excessive size shall not be furnished.

3.6.2 Cast-to-size parts shall conform to dimensions and tolerances specified and shall not be warped or distorted in such a manner as to adversely affect their appearance or serviceability.

3.7 Marking.

3.7.1 Small castings that are impracticable to mark individually shall be properly separated by melts and so marked that their identity will be clear at the point of delivery.

3.7.2 Large castings shall be marked with the pattern, mark, drawing, piece number, and any other data as specified in the contract or order (see 6.1), and, when practicable, with the melt or lot number, in a position on the casting where they will not be machined off in manufacturing to finished dimensions.

3.8 Workmanship. Castings shall be of uniform quality and condition, free from checks, pits, blowholes, porosity, hard spots, shrinkage defects, cracks, or any other defects which are detrimental to their intended use, or are revealed by manufacturing operations performed subsequent to acceptance (see 6.4). Castings shall be of the highest grade of workmanship and shall meet all the requirements of the specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For the purpose of sampling, a lot shall consist of all the castings produced from one pouring of a furnace, crucible, or a ladle. When two or more furnace or crucible melts are used to charge a ladle for a pouring, the castings produced therefrom shall comprise the lot for sampling purposes. Castings shall be grouped by lots or melts when presented for inspection.

4.2.2 Sampling for chemical analysis. A sample for chemical analysis shall be taken from tension test specimens or from a representative casting from each lot. The drillings shall be taken from sound metal below the surface in accordance with method 111 of FED-STD-151. The sample shall be not less than 2 ounces of drillings.

4.2.3 Sampling for mechanical properties. A tension test sample shall be cut from representative castings of each lot or from bars of the same melt, same cross section, and heat treatment as the castings of each lot.

4.2.3.1 Specimens. Tensile test specimens shall conform to the requirements of ASTM E 8. The specimen type with the largest diameter that can be machined from the sample shall be used for test.

4.2.3.2 Location. For castings up to 1-1/2 inches in diameter or thickness, the axis of the tensile test specimen shall coincide with the central axis of the castings; for castings 1-1/2 inches and over in diameter or thickness, the axis shall be midway between the center and surface of the casting. The axis of the tensile test specimen shall be parallel to the major axis of the casting.

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4.3 Inspection.

4.3.1 Surface. The surface of each casting shall be inspected to determine conformance to the requirements of 3.4, 3.5, 3.7, and 3.8. Any casting which fails to conform to these requirements shall be rejected.

*4.3.2 Dimensional.

4.3.2.1 A representative sample from each lot of castings shall be inspected to determine compliance with the required dimensions and tolerances.

4.4 Chemical analysis. The sample obtained in accordance with 4.2.2 shall be analyzed in accordance method 111 of FED-STD-151 to determine conformance to the requirements of 3.2. If the sample fails to conform to the requirements, the lot represented by the sample shall be rejected.

4.5 Tension tests. Tension tests shall be conducted in accordance with method ASTM E 8. The yield strength shall be determined by the extension-under-load methods.

5. PREPARATION FOR DELIVERY

5.1 General.

5.1.1 Unless otherwise specified (see 6.1), all castings shall be separated by size, shape, or pattern when packed for shipment.

5.1.2 Where practicable, shipping containers shall be of uniform size, and shall contain the identical number of castings of the same pattern. Containers shall be designed to fit the contents in a compact manner. Castings shall be adequately blocked, braced, or otherwise secured to prevent their movement within the shipping containers.

5.2 Packaging. Packaging shall be levels A or C, as specified (see 6.1).

5.2.1 Level A. Unless otherwise specified (see 6.1), castings having a finished or polished surface shall be packaged in accordance with

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MIL-P-116, method III. When a PPP-B-636, class weather-resistant container is used to supplement the packaging method, closure may be accomplished using two strips of tape conforming to PPP-T-76. The tape will be 2 inches in width and will be applied to top and bottom. If the bottom is closed with staples or adhesive (or a style OPF container is used), only a single piece of tape on the top seam is required. Tape need not extend over end panel.

5.2.1.1 Rough castings not subject to damage do not require packaging.

5.2.2 Level C. Castings shall be packaged in accordance with the manufacturer's commercial practice.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.3.1 Level A. Castings packaged as specified in 5.2.1 shall be packed in overseas type containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636 (class weather-resistant), PPP-B-576, or PPP-C-650. Insofar as practical, containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Castings having a finished or polished surface subject to damage shall be afforded adequate blocking or bracing within the shipping container. When specified, wood and wood cleated containers shall be provided with a case liner conforming to MIL-L-10547, and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the unit container conforms to PPP-B-636, class weather resistant, or the exterior container conforms to PPP-B-636, class weather resistant, and is sealed in accordance with the container specification requirements. Containers shall be closed and strapped in accordance with applicable container specifications or appendix thereto. The gross weight and size shall not exceed the limitation of the applicable container specification.

5.3.1.1 Unless otherwise specified (see 6.1), rough castings shall be shipped loose or in bundles or palletized loads. Bundles not exceeding 500 pounds maximum shall be securely strapped with a minimum of 1 1/4-inch wide strapping conforming to QQ-S-781, type I. Palletized loads shall be in accordance with MIL-STD-147.

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5.3.2 Level B. Castings packaged as specified in 5.2.1 shall be packed in domestic type containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636, PPP-B-576, or PPP-C-650. Insofar as practical, containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Castings having a finished or polished surface subject to damage shall be afforded adequate blocking or bracing within the shipping containers. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. The gross weight and size shall not exceed the limitation of the applicable container specification.

5.3.2.1 Unless otherwise specified (see 6.1), large rough castings shall be packed as specified in 5.3.1.1.

5.3.3 Level C. Castings shall be packed for shipment in compliance with common carrier regulations applicable to that mode of transportation to ensure safe delivery at destination at lowest transportation cost without assessment of penalty charges for improper packing.

5.4 Marking.

5.4.1 Civilian agencies. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with FED-STD-123.

5.4.2 Military agencies. In addition to any special marking required in the contract or order, marking of the shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Whether patterns are to be furnished by the Government or are to be supplied by the contractors

(c) Dimensions required (see 3.6)

(d) Markings required (see 3.7)

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(e) Levels of packaging and packing required (see 5.2 and 5.3)

(f) Special marking, if any (see 5.4).

6.2 Patterns. All patterns furnished by the Government shall be properly cared for, and, when being returned, shall be suitably boxed or crated by the contractor to prevent damage or breakage. The patterns shall be examined by the Government inspector before shipment and report made if found in unsatisfactory condition.

6.3 Replacement of defective castings. The contractor may be required to replace castings in which injurious defects are revealed by manufacturing operations performed subsequent to acceptance, without expense to the Government.

*6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing activity:
Navy - OS
(Project No. MECA-N041)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-17112A (OS), Copper-Nickel-Zinc Alloy (Nickel-Silver): Castings

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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